

Design and Implementation of a Smart Multi-Speciality Hospital Network Using Cisco Packet Tracer

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LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course
**CSE322: Computer Networks Lab in the Department of Computer
Science and Engineering**



DAFFODIL INTERNATIONAL UNIVERSITY

Dhaka, Bangladesh

11/08/2026

DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Mr. Tanvirul islam , Lecturer** Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:

Table 1: Course Outcome Statements

CO's	Statements
CO1	Describe and apply the fundamentals of LAN architecture along with basic topology setup using simulation tools.
CO2	Configure and Analyze network devices and services using Cisco Packet Tracer in various networking topologies.
CO3	Design, Implement, and Troubleshoot dynamic and static routing protocols NAT, and ACLs in simulated environments to build efficient and secure networks.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP	CEA
CO1	PO5	C3, P1, P2, A2	K6	EP1	
CO2	PO2	C3, P2	K1-K4	EP2	
CO3	PO3	A2, P2	K5	EP3	

The mapping justification of this table is provided in section 4.3.1, 4.3.2 and 4.3.3.

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Chapter 1

Introduction

This chapter introduces the proposed multi-site hospital network and defines the networking problem addressed by the project. It presents the motivation, objectives, and expected outcomes while establishing the engineering considerations involved in designing and implementing the network infrastructure.

1.1 Introduction

Modern healthcare organizations depend on reliable and well-structured computer networks to support communication, resource sharing, Internet access, and information exchange among different departments and locations. In a multi-site hospital environment, the network must provide reliable connectivity while maintaining appropriate security, scalability, accessibility, and efficient resource utilization.

This project focuses on the **design and implementation of a multi-site hospital network infrastructure using Cisco Packet Tracer**. The proposed network represents a healthcare organization consisting of a Main Hospital Headquarters and five additional hospital branches. The network is designed to provide communication among different sites and departments while supporting required network and Internet-based services. The project applies fundamental computer networking concepts, including network topology, IP addressing, subnetting, routing, network device configuration, and simulation-based network design. These concepts are translated into practical configurations using routers, switches, servers, access points, and end devices in Cisco Packet Tracer.



Figure 1: Healthcare Network Applications

The proposed network incorporates **VLSM-based IP addressing, VLAN segmentation, Router-on-a-Stick inter-VLAN routing, DHCP, EIGRP dynamic routing, NAT/PAT**, to address the functional and security requirements of the hospital environment. The integration of these components requires appropriate consideration of accessibility, security, scalability, reliability, and resource utilization. The network consists of multiple interacting components, including routers, switches, servers, end devices, subnets, routing protocols, NAT, and ACLs. Therefore, the project requires analysis of network behavior, packet paths, routing information, connectivity, and security policies to ensure that the selected design satisfies the required networking objectives.

The complete network is implemented and simulated in Cisco Packet Tracer. Connectivity, routing, NAT, ACL behavior, and end-to-end communication are tested to identify configuration problems and validate the proposed solution. This practical implementation demonstrates the application of computer networking knowledge to a realistic and complex engineering problem.

1.2 Motivation

The motivation of this project is to develop a **reliable, secure, and scalable network infrastructure** for a multi-site hospital environment. The network is intended to support continuous communication and resource sharing among different hospital departments and branches.

- To ensure reliable communication between different hospital departments and branches.
- To provide secure and controlled access to network resources.
- To develop a scalable network that can support future users, devices, departments, and services.
- To apply networking concepts such as **IP addressing, subnetting, VLAN, routing, DHCP, NAT**, in a practical environment.
- To gain practical experience in network design, configuration, testing, and troubleshooting using **Cisco Packet Tracer**.
- To understand the interaction between routers, switches, servers, end devices, routing protocols, and security mechanisms.
- To develop an efficient network solution while considering **security, accessibility, reliability, scalability, and resource utilization**.

Overall, this project aims to develop practical networking skills through the design, implementation, testing, and troubleshooting of a realistic multi-site hospital network infrastructure.

1.3 Objectives

The main objectives of this project are:

- To design a **multi-site hospital network** connecting the main hospital and its branches.
- To develop an efficient **IP addressing and VLSM-based subnetting scheme**.
- To implement **VLANs** for proper departmental network segmentation.
- To configure **inter-VLAN routing** using Router-on-a-Stick.
- To implement **DHCP** for automatic IP address assignment.
- To configure **dynamic routing using EIGRP** for communication between different network sites.
- To implement **NAT/PAT** for Internet connectivity.
- To configure **ACLs** to control network traffic and improve security.
- To configure required **network services and end devices** in Cisco Packet Tracer.
- To test **end-to-end connectivity, routing, NAT** behavior.
- To analyze and troubleshoot network problems using appropriate testing and verification methods.
- To develop a **reliable, secure, scalable, and manageable** network infrastructure.

1.4 Feasibility Study

The proposed multi-site hospital network is feasible from **technical, operational, and resource perspectives**. The required network infrastructure can be designed and implemented using Cisco Packet Tracer, which provides routers, switches, servers, access points, and end devices required to represent the proposed hospital environment.

The feasibility of the proposed multi-site hospital network is evaluated from the following perspectives:

1. **Technical Feasibility:** The proposed network can be implemented in Cisco Packet Tracer using routers, switches, servers, access points, and end devices. The simulation environment supports the required configurations and network services.
2. **Network Feasibility:** VLSM, VLAN, Router-on-a-Stick, DHCP, EIGRP, NAT/PAT can be integrated to support addressing, segmentation, routing, Internet connectivity, and controlled network access.
3. **Operational Feasibility:** The proposed design supports communication between hospital departments and branches while maintaining departmental separation and controlled access to network resources.
4. **Security Feasibility:** VLAN segmentation can be used to restrict unauthorized traffic while allowing required communication between users, departments, and services.
5. **Scalability Feasibility:** The proposed architecture can accommodate additional users, devices, departments, and branches by extending the existing addressing, VLAN, and routing structure.
6. **Cost Feasibility:** Cisco Packet Tracer allows the complete network to be designed, configured, tested, and troubleshooted without requiring physical networking equipment during the development stage.
7. **Testing Feasibility:** Connectivity, routing, NAT, and packet flow can be verified through ping, routing-table inspection, and Packet Tracer simulation tools.

Therefore, the proposed network is technically implementable, operationally suitable, secure, scalable, and cost-effective for the project requirements.

1.5 Gap Analysis

Many basic networking projects focus on single-site networks with limited devices, simple addressing, and basic connectivity. Such implementations do not fully represent the challenges of a multi-site healthcare environment where multiple departments, branches, routing domains, security policies, and network services must operate together. This project addresses this gap by developing a **six-site hospital network** with department-level VLAN segmentation and VLSM-based addressing. The design integrates dynamic routing using EIGRP, DHCP, NAT/PAT based access control, and multiple network services within the same infrastructure. The project also goes beyond basic connectivity by analyzing **routing behavior, packet paths, NAT translation, and network troubleshooting**. This provides a more realistic representation of the design and analysis required for a complex multi-site network.

1.6 Project Outcome

The project is expected to achieve the following outcomes:

- A **functional multi-site hospital network** connecting the Main Hospital and five branch sites.
- Efficient IP address allocation using **VLSM-based subnetting**.
- Proper departmental network segmentation using **VLANs**.
- Successful **inter-VLAN and inter-branch communication**.
- Automatic IP address assignment through **DHCP**.
- Dynamic routing between sites using **EIGRP**.
- Internet connectivity through **NAT/PAT**.
- Verification of connectivity using **ping**
- Analysis of routing tables, NAT translations behavior.
- A **reliable, secure, scalable, and manageable** network architecture.

Chapter 2

Network Architecture

This chapter describes the overall architecture and design of the proposed **Smart Multi-Speciality Hospital Network**. It presents the network requirements, topology, IP addressing and subnetting scheme, and the simulation tools used to implement, test, and validate the network.

2.1 Requirement Analysis & Design Specification

2.1.1 Requirements

The proposed **Smart Multi-Speciality Hospital Network** requires a combination of networking devices, end devices, network services, and security mechanisms to provide reliable communication among the six hospital sites.

Table 2.1 :Hardware and Network

Requirement	Quantity
Hospital Sites	6
Routers	13
Core Routers	6
Edge Routers	6
ISP Router	1
Layer 3 Core Switch	1
Access Switches	6
Servers	5
PCs / Laptops	23
Internet/ISP Connection	1
Access Points	4
Printers	5
Tablets / Smartphones	8

Table 2.2: Network and Configuration Requirements

Requirement	Specification
IP Addressing	VLSM-based IPv4 addressing
Network Segmentation	VLAN-based departmental segmentation
Inter-VLAN Routing	Router-on-a-Stick
Dynamic Routing	EIGRP, AS 100
IP Assignment	DHCP
Internet Connectivity	NAT/PAT
Network Security	Access Control Lists (ACLs)
Network Services	DNS, HTTP, FTP, SMTP/Email
Connectivity Testing	Ping
Packet Analysis	Cisco Packet Tracer Simulation Mode
Simulation Platform	Cisco Packet Tracer

In Table 1, the network consists of six hospital sites: Main Hospital HQ, Diagnostic Center, Emergency Clinic, Telemedicine, Maternity Hospital, and Children Care Hospital. Each site uses an Edge Router and a Core Router. The documented configuration identifies the ISP router as R0, with Edge Routers R1, R4, R6, R9, R11, and R13, and Core Routers R2, R5, R7, R10, R12, and R14. In Table 2, the network uses VLAN-based segmentation for different departments. For example, the Main Hospital HQ contains VLAN 10 for DMZ, VLAN 20 for Doctors & ICU, VLAN 30 for Billing & Admin, and VLAN 50 for Guest Wi-Fi. EIGRP is configured using Autonomous System 100 with no auto-summary across the hospital network. For Internet connectivity, NAT/PAT is configured on the Edge Routers using overload. Inter-branch traffic within the internal 10.0.0.0/8 address space is excluded from NAT so that branch-to-branch communication can occur through the routing infrastructure without unnecessary address translation.

2.1.2 Network Topology Diagram

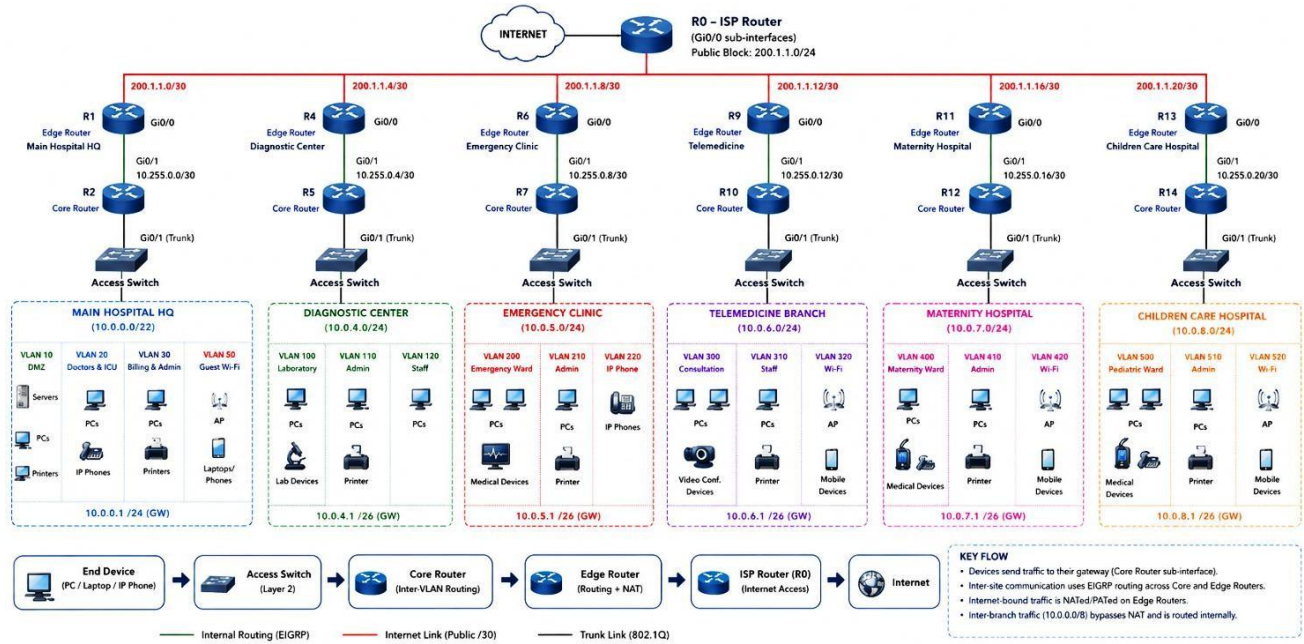


Fig 2.1 :Network Topology Diagram

The system architecture illustrates the overall connectivity and communication flow of the proposed Smart Multi-Speciality Hospital Network. The network consists of six hospital sites, where each site is connected through an **Edge Router and Core Router**. The Edge Routers provide connectivity toward the ISP, while the Core Routers connect the departmental Layer 2 switches through VLAN-based segmentation. Inter-VLAN communication is enabled using **Router-on-a-Stick**, while **EIGRP** provides dynamic routing between the hospital sites. Internet access is provided through **NAT/PAT** at the Edge Routers, allowing internal hospital networks to securely communicate with external networks.

2.1.3 IP Addressing & Subnetting Table

This table presents the IP addressing configuration for the point-to-point connections between the ISP Router (R0) and the Edge Routers of the six hospital sites. Each connection uses a /30 subnet from the **200.1.1.0/24** address space, providing dedicated WAN connectivity between the ISP and each hospital branch.

Table 2.3: ISP-Edge Router Link IP Addressing

Site	Router	Interface	IP Address	Prefix
Main Hospital HQ	R0	Gi0/0.101	200.1.1.2	/30
Main Hospital HQ	R1 (Edge)	Gi0/0	200.1.1.1	/30
Diagnostic Center	R0	Gi0/0.104	200.1.1.6	/30
Diagnostic Center	R4 (Edge)	Gi0/0	200.1.1.5	/30
Emergency Clinic	R0	Gi0/0.106	200.1.1.10	/30
Emergency Clinic	R6 (Edge)	Gi0/0	200.1.1.9	/30
Telemedicine	R0	Gi0/0.109	200.1.1.14	/30
Telemedicine	R9 (Edge)	Gi0/0	200.1.1.13	/30
Maternity Hospital	R0	Gi0/0.111	200.1.1.18	/30
Maternity Hospital	R11 (Edge)	Gi0/0	200.1.1.17	/30
Children Care Hospital	R0	Gi0/0.113	200.1.1.22	/30
Children Care Hospital	R13 (Edge)	Gi0/0	200.1.1.21	/30

This table presents the IP addressing configuration for the point-to-point connections between the Edge Router and Core Router at each of the six hospital sites. Each Edge–Core connection uses a **/30 subnet** from the **10.255.0.0/24** address space, providing dedicated internal connectivity between the Edge and Core layers of each hospital network.

Table 2.4: Edge Router–Core Router Link IP Addressing

Site	Edge Router (Interface)	Edge IP	Core Router (Interface)	Core IP	Subnet
Main Hospital HQ	R1 (Gi0/1)	10.255.0.1	R2 (Gi0/0)	10.255.0.2	/30
Diagnostic Center	R4 (Gi0/1)	10.255.0.5	R5 (Gi0/0)	10.255.0.6	/30
Emergency Clinic	R6 (Gi0/1)	10.255.0.9	R7 (Gi0/0)	10.255.0.10	/30
Telemedicine	R9 (Gi0/1)	10.255.0.13	R10 (Gi0/0)	10.255.0.14	/30
Maternity Hospital	R11 (Gi0/1)	10.255.0.17	R12 (Gi0/0)	10.255.0.18	/30
Children Care Hospital	R13 (Gi0/1)	10.255.0.21	R14 (Gi0/0)	10.255.0.22	/30

Table 2.5 presents the VLAN-based network segmentation for different departments across the six hospital sites. It includes the VLAN ID, VLAN name, sub-interface, assigned switch ports, subnet size, network address, prefix, default gateway, and number of usable hosts.

Table 2.5: VLAN Configuration and Subnet Allocation

Site / Router	VLAN ID	VLAN Name	Sub-Interface	Port Range	Block Size	Network Address	Pr efi	Default Gateway	Usable Hosts
Main Hospital HQ	10	DMZ	Gi0/1.10	Fa0/1–5	16	10.0.1.160	/28	10.0.1.161	14
Main Hospital HQ	20	DOCTORS_ICU	Gi0/1.20	Fa0/6–10	64	10.0.1.64	/26	10.0.1.65	62
Main Hospital HQ	30	BILLING_ADMIN	Gi0/1.30	Fa0/11–15	64	10.0.1.0	/26	10.0.1.1	62
Main Hospital HQ	50	GUEST_WIFI	Gi0/1.50	Fa0/16	256	10.0.0.0	/24	10.0.0.1	254
Diagnostic Center	100	LAB	Gi0/1.100	Fa0/1–10	64	10.1.0.0	/26	10.1.0.1	62
Diagnostic Center	110	ADMIN	Gi0/1.110	Fa0/11–15	32	10.1.0.64	/27	10.1.0.65	30
Diagnostic Center	120	STAFF	Gi0/1.120	Fa0/16–20	16	10.1.0.96	/28	10.1.0.97	14
Emergency Clinic	200	EMERGENCY	Gi0/1.200	Fa0/1–10	64	10.2.0.0	/26	10.2.0.1	62
Emergency Clinic	210	ADMIN	Gi0/1.210	Fa0/11–15	32	10.2.0.64	/27	10.2.0.65	30
Emergency Clinic	220	IP_PHONE	Gi0/1.220	Fa0/16–20	16	10.2.0.96	/28	10.2.0.97	14
Telemedicine	300	CONSULTATION	Gi0/1.300	Fa0/1–10	64	10.3.0.0	/26	10.3.0.1	62
Telemedicine	310	ADMIN	Gi0/1.310	Fa0/11–15	32	10.3.0.64	/27	10.3.0.65	30
Telemedicine	320	WIFI	Gi0/1.320	Fa0/16	32	10.3.0.128	/27	10.3.0.129	30
Maternity Hospital	400	MATERNITY	Gi0/1.400	Fa0/1–10	64	10.4.0.0	/26	10.4.0.1	62
Maternity Hospital	410	ADMIN	Gi0/1.410	Fa0/11–15	32	10.4.0.64	/27	10.4.0.65	30
Maternity Hospital	420	WIFI	Gi0/1.420	Fa0/16	32	10.4.0.128	/27	10.4.0.129	30
Children Care Hospital	500	CHILDREN	Gi0/1.500	Fa0/1–10	64	10.5.0.0	/26	10.5.0.1	62
Children Care Hospital	510	ADMIN	Gi0/1.510	Fa0/11–15	32	10.5.0.64	/27	10.5.0.65	30
Children Care Hospital	520	WIFI	Gi0/1.520	Fa0/16	32	10.5.0.128	/27	10.5.0.129	30

Table 2.6 summarizes the NAT/PAT configuration implemented on the Edge Routers to provide Internet access for each hospital site. Internal traffic destined for the **10.0.0.0/8** network is excluded from NAT, while external traffic is translated using **PAT overload** through the Edge Router's **Gi0/0** interface.

Table 2.6: NAT/PAT Configuration Summary

Router	Site	Own Subnet	ACL Deny (No NAT)	ACL Permit (NAT)	NAT Interface
R1	HQ Edge	10.0.0.0/22	10.0.0.0/22 → 10.0.0.0/8	10.0.0.0/22 → any	Gi0/0 (overload)
R4	Diagnostic Center Edge	10.1.0.0/24	10.1.0.0/24 → 10.0.0.0/8	10.1.0.0/24 → any	Gi0/0 (overload)
R6	Emergency Clinic Edge	10.2.0.0/24	10.2.0.0/24 → 10.0.0.0/8	10.2.0.0/24 → any	Gi0/0 (overload)
R9	Telemedicine Edge	10.3.0.0/24	10.3.0.0/24 → 10.0.0.0/8	10.3.0.0/24 → any	Gi0/0 (overload)
R11	Maternity Hospital Edge	10.4.0.0/24	10.4.0.0/24 → 10.0.0.0/8	10.4.0.0/24 → any	Gi0/0 (overload)
R13	Children C Hospital Edge	10.5.0.0/24	10.5.0.0/24 → 10.0.0.0/8	10.5.0.0/24 → any	Gi0/0 (overload)

2.2 Use of Modern Simulation Tools

In Table 2.7, the modern simulation tools and networking technologies used to design, configure, simulate, and verify the proposed Smart Multi-Speciality Hospital Network are presented. Cisco Packet Tracer was used as the primary simulation platform, while VLAN, VLSM, EIGRP, NAT/PAT, DHCP, and connectivity-testing tools were used to implement and evaluate different network functions.

Table 2.7: Modern Simulation Tools and Networking Technologies Used

Tool	Purpose	Application in the Project
Cisco Packet Tracer	Network design and simulation	Designed and simulated the complete hospital network topology
Router Configuration	Network connectivity	Configured ISP, Edge, and Core Routers
Switch Configuration	LAN management	Configured switches and departmental network connections
VLAN	Network segmentation	Separated different hospital departments into individual VLANs
VLSM	Efficient IP addressing	Allocated appropriate IP address ranges according to departmental requirements
EIGRP	Dynamic routing	Enabled dynamic route exchange among the hospital networks
NAT/PAT	Internet connectivity	Provided Internet access through Edge Routers using PAT overload
DHCP	Automatic IP assignment	Assigned IP addresses automatically to network devices
Simulation Mode	Network verification	Tested packet flow and communication between different network segments
Ping	Connectivity testing	Verified end-to-end connectivity and routing paths

Chapter 3

Implementation and Results

This chapter presents the detailed implementation and evaluation of the proposed **Smart Multi-Speciality Hospital Network** using Cisco Packet Tracer. It describes the configuration of routers, switches, VLANs, IP addressing, routing protocols, network services, NAT/PAT, and other required components. The implementation is supported by configuration commands and screenshots of the Cisco Packet Tracer environment. The output section presents the results obtained from different connectivity and service tests, including HTTP, FTP, DHCP, inter-VLAN communication, EIGRP routing, NAT/PAT, and end-to-end connectivity.

3.1 Implementation

This section describes the step-by-step implementation of the proposed hospital network in Cisco Packet Tracer. The implementation includes the physical and logical topology, device configuration, IP addressing, VLAN configuration, inter-VLAN routing, dynamic routing, network services, and Internet connectivity.

Project: <https://github.com/arpi333/Hospital-Network-Design-Cisco>

3.1.1 Network Topology Implementation

The complete network topology was implemented in Cisco Packet Tracer according to the proposed system architecture. The network consists of six hospital sites connected through an ISP router, with each site containing an Edge Router and a Core Router. Switches and end devices are connected to the Core Router according to their respective departmental requirements.

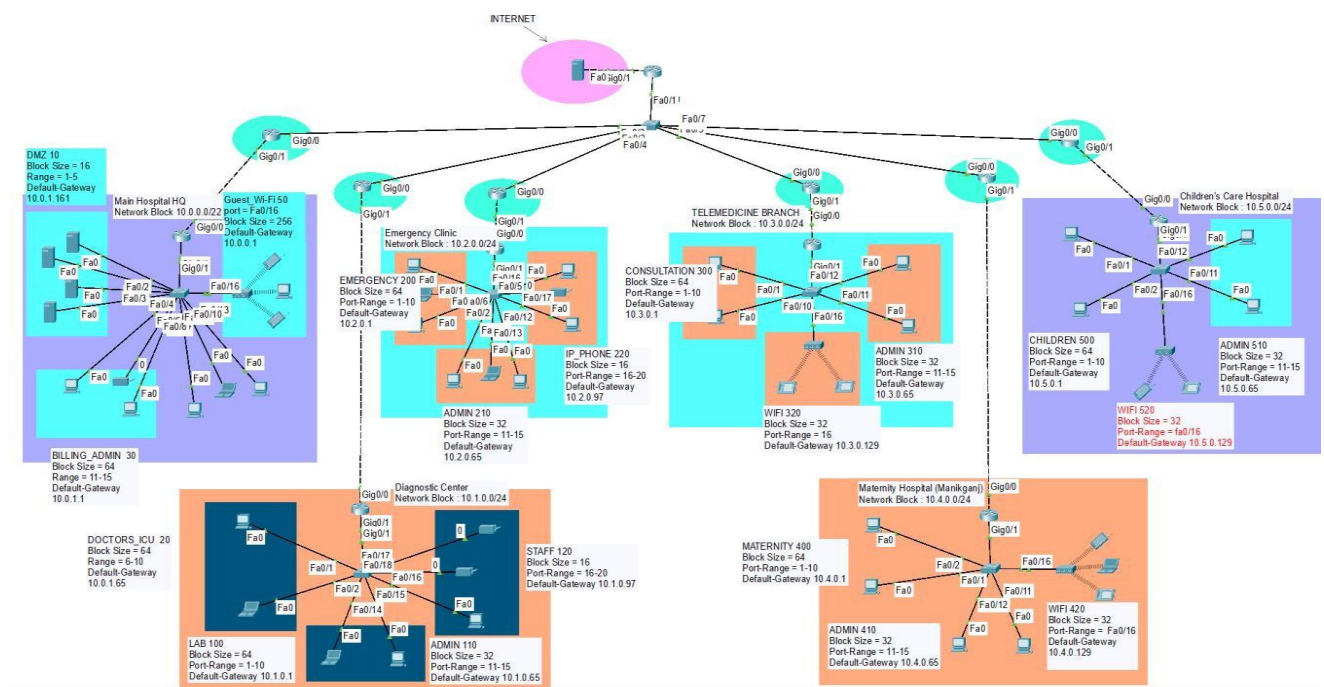


Figure 3.1: Complete Cisco Packet Tracer Network Topology

3.1.2 Main Hospital HQ Topology

The Main Hospital HQ topology was implemented according to the proposed network design. It includes the Edge Router, Core Router, switch, departmental VLANs, servers, and connected end devices.

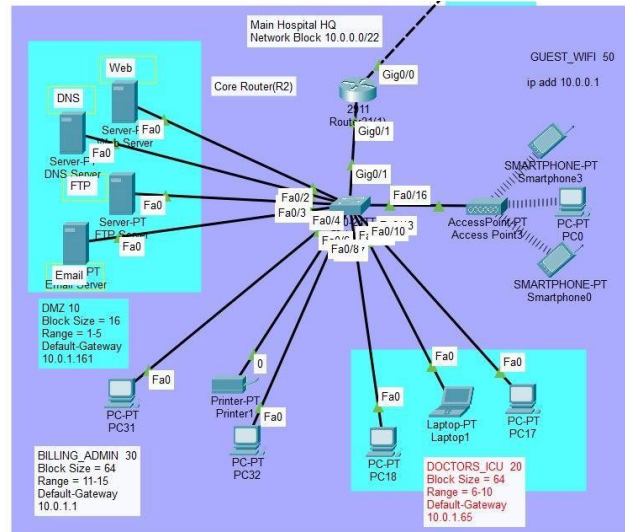


Figure 3.2: Main Hospital HQ Network Topology

3.1.3 Diagnostic Center Topology

The Diagnostic Center topology was implemented with the required Edge Router, Core Router, switch, and departmental network segments. The Laboratory, Admin, and Staff networks were connected according to the designed topology.

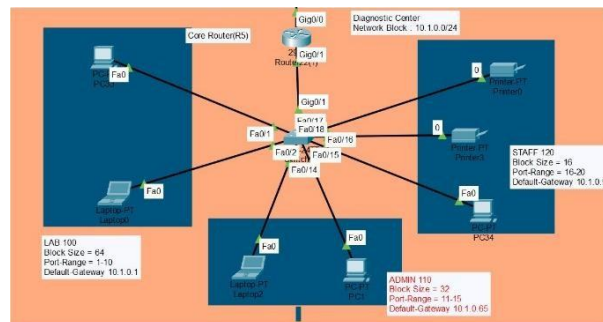


Figure 3.3: Diagnostic Center Network Topology

3.1.4 Emergency Clinic Topology

The Emergency Clinic topology was implemented with the required routing and switching devices. The Emergency Ward, Admin, and IP Phone networks were connected according to the proposed design.

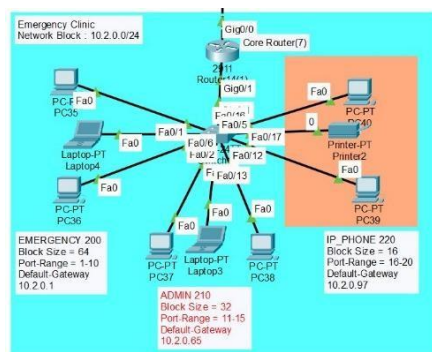


Figure 3.4: Emergency Clinic Network Topology

3.1.8 Router Configuration

The ISP, Edge, and Core Routers were configured according to the designed IP addressing scheme. The router interfaces were assigned appropriate IP addresses and subnet masks to establish communication between the different network segments.

Main Hospital HQ Edge Router (R1)

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 200.1.1.1 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.255.0.1 255.255.255.252
no shutdown
exit
```

Main Hospital HQ Core Router (R2)

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 10.255.0.2 255.255.255.252
no shutdown
exit
```

3.1.9 VLAN Configuration

VLANs were configured on the access switches to logically separate different departments within each hospital site. Each department was assigned a specific VLAN ID and name according to the designed VLAN and subnet allocation. The required switch ports were configured as access ports and assigned to their respective VLANs. For example, the **Main Hospital HQ** uses VLAN 10 for DMZ, VLAN 20 for Doctors & ICU, VLAN 30 for Billing & Admin, and VLAN 50 for Guest Wi-Fi.

Switch Configuration

```
en
conf t

vlan 10
name DMZ

vlan 20
name DOCTORS_ICU

vlan 30
name BILLING_ADMIN

vlan 50
name GUEST_WIFI

exit

int range fa0/1-5
switchport mode access
switchport access vlan 10
exit

int range fa0/6-10
switchport mode access
switchport access vlan 20
exit

int range fa0/11-15
switchport mode access
switchport access vlan 30
exit

int fa0/16
switchport mode access
switchport access vlan 50
exit

int g0/1

switchport mode trunk
switchport trunk allowed vlan 10,20,30,50
no shut
exit
```

Figure 3.8: VLAN Configuration on Switch

Router Configuration

```

en
conf t

int g0/1
no shut
exit

int g0/1.10
encapsulation dot1Q 10
ip add 10.0.1.161 255.255.255.240
exit

int g0/1.20
encapsulation dot1Q 20
ip add 10.0.1.65 255.255.255.192
exit

int g0/1.30
encapsulation dot1Q 30
ip add 10.0.1.1 255.255.255.192
exit

int g0/1.50
encapsulation dot1Q 50
ip add 10.0.0.1 255.255.255.0
exit

ip dhcp excluded-address 10.0.1.161

ip dhcp pool DMZ
network 10.0.1.160 255.255.255.240
default-router 10.0.1.161
dns-server 10.0.1.162
exit

ip dhcp excluded-address 10.0.1.65

ip dhcp pool DOCTORS_ICU
network 10.0.1.64 255.255.255.192
default-router 10.0.1.65
dns-server 10.0.1.162
exit

```

Figure 3.9: VLAN Configuration on Router

3.1.10 EIGRP Configuration

EIGRP was configured to provide dynamic routing between the hospital networks. **Autonomous System 100** was used throughout the network, with no auto-summary enabled. Example configuration:

R1 (HQ Edge)

```

en
conf t

router eigrp 100
no auto-summary

network 200.1.1.0 0.0.0.3
network 10.255.0.0 0.0.0.3

exit

```

R2 (HQ Core)

```

en
conf t

router eigrp 100
no auto-summary

network 10.255.0.0 0.0.0.3
network 10.0.0.0 0.0.3.255

exit

```

R4 (Diagnostic Edge)

```

en
conf t

router eigrp 100
no auto-summary

network 200.1.1.4 0.0.0.3
network 10.255.0.4 0.0.0.3

exit

```

R5 (Diagnostic Core)

```

en
conf t

router eigrp 100
no auto-summary

network 10.255.0.4 0.0.0.3
network 10.1.0.0 0.0.0.255

exit

```

R11 (Maternity Edge)

```

en
conf t

router eigrp 100
no auto-summary

network 200.1.1.16 0.0.0.3
network 10.255.0.16 0.0.0.3

exit

```

R12 (Maternity Core)

```

en
conf t

router eigrp 100
no auto-summary

network 10.255.0.16 0.0.0.3
network 10.4.0.0 0.0.0.255

exit

```

Figure 3.10: EIGRP Configuration and Routing Table

3.1.11 DHCP Configuration

DHCP was configured to automatically assign IP addresses to end devices within the required VLANs.

```

ip dhcp excluded-address 10.0.1.161
ip dhcp pool DMZ
network 10.0.1.160 255.255.255.240

```

```
default-router 10.0.1.161
dns-server 10.0.1.162
exit
```

```
ip dhcp excluded-address 10.0.1.65
ip dhcp pool DOCTORS_ICU
network 10.0.1.64 255.255.255.192
default-router 10.0.1.65
dns-server 10.0.1.162
exit
```

```
ip dhcp excluded-address 10.0.1.1
ip dhcp pool BILLING_ADMIN
network 10.0.1.0 255.255.255.192
default-router 10.0.1.1
dns-server 10.0.1.162
exit
```

```
ip dhcp excluded-address 10.0.0.1
ip dhcp pool GUEST_WIFI
network 10.0.0.0 255.255.255.0
default-router 10.0.0.1
dns-server 8.8.8.8
exit
```

3.1.12 NAT/PAT Configuration

NAT/PAT was configured on the Edge Routers to provide Internet connectivity to the internal hospital networks. PAT overload allows multiple internal devices to share the available external IP address. The ACL configuration also excludes internal **10.0.0.0/8** traffic from NAT so that communication between hospital branches remains within the internal routing infrastructure.

R1 — HQ Edge (10.0.0.0/22)

```
en
conf t

no ip nat inside source list 1 interface g0/0 overload
no access-list 1

ip access-list extended NAT_ACL
deny ip 10.0.0.0 0.0.3.255 10.0.0.0 0.255.255.255
permit ip 10.0.0.0 0.0.3.255 any
exit

int g0/1
ip nat inside
exit

int g0/0
ip nat outside
exit

ip nat inside source list NAT_ACL interface g0/0 overload
end
```

R6 — Emergency Clinic Edge (10.2.0.0/24)

```
en
conf t

ip access-list extended NAT_ACL
deny ip 10.2.0.0 0.0.0.255 10.0.0.0 0.255.255.255
permit ip 10.2.0.0 0.0.0.255 any
exit

int g0/1
ip nat inside
exit

int g0/0
ip nat outside
exit

ip nat inside source list NAT_ACL interface g0/0 overload
end
```

Figure 3.11: NAT/PAT Configuration

3.1.13 Network Services Configuration

The required servers and network services were configured in Cisco Packet Tracer according to the proposed hospital network design. The configured services include **HTTP, FTP, DNS, and other services available in the implemented network**. The server configurations were performed to provide centralized network services to the connected hospital users and departments. The configuration and activation status of each service were verified before performing the output tests.

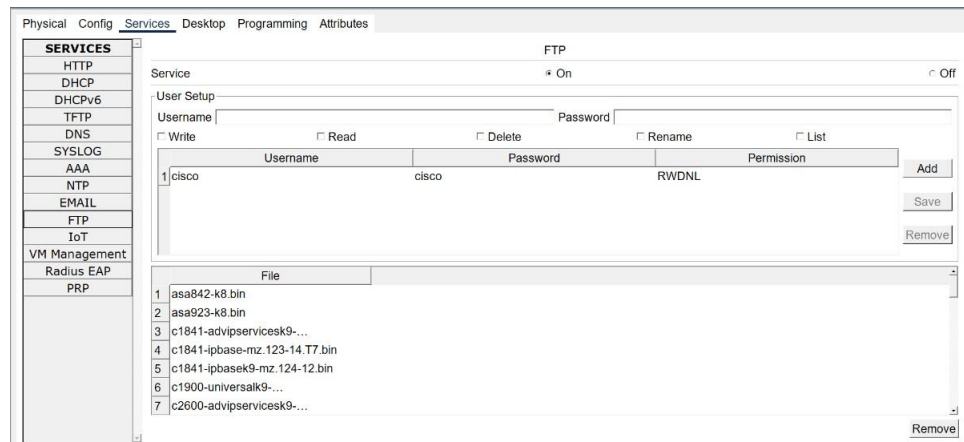


Figure 3.12: Server Configuration and Enabled Services

3.1.14 End Device Configuration

The end devices connected to the hospital network were configured according to their respective VLANs and network requirements. This includes PCs, laptops, IP phones, printers, and other end devices present in the implemented topology. The devices were configured with appropriate network settings, either manually or through DHCP, to enable communication with their respective gateways, servers, and other network segments.

3.2 Output

This chapter presents the implementation and testing of the proposed Smart Multi-Speciality Hospital Network using Cisco Packet Tracer. It describes the configuration of the network devices, VLANs, IP addressing, DHCP, routing protocols, NAT/PAT, and network services. The chapter also presents the output obtained from each hospital branch and the results of different connectivity and service tests.

3.2.1 Basic Connectivity Test

Basic connectivity was tested using the **Ping** command between different network devices and end devices. Successful responses confirm that the assigned IP addresses, subnet masks, default gateways, and physical connections are functioning correctly.

```
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.1.64

Pinging 10.0.1.64 with 32 bytes of data:

Reply from 10.255.0.2: bytes=32 time=36ms TTL=251
Reply from 10.255.0.2: bytes=32 time=3ms TTL=251
Reply from 10.255.0.2: bytes=32 time=1ms TTL=251
Reply from 10.255.0.2: bytes=32 time<1ms TTL=251

Ping statistics for 10.0.1.64:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 36ms, Average = 10ms
C:\>
```

```
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.1.65

Pinging 10.0.1.65 with 32 bytes of data:

Reply from 10.0.1.65: bytes=32 time=37ms TTL=251
Reply from 10.0.1.65: bytes=32 time=17ms TTL=251
Reply from 10.0.1.65: bytes=32 time=11ms TTL=251
Reply from 10.0.1.65: bytes=32 time=5ms TTL=251

Ping statistics for 10.0.1.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 37ms, Average = 17ms
```

Figure 3.13: Successful Basic Connectivity Test

3.2.2 Inter-VLAN Communication Test

Communication between different departmental VLANs was tested to verify the Router-on-a-Stick configuration. Successful communication confirms that the VLANs, trunk connection, and Core Router sub-interfaces are properly configured.

```

C:\>ping 10.0.1.13

Pinging 10.0.1.13 with 32 bytes of data:

Reply from 10.0.1.13: bytes=32 time<1ms TTL=123
Reply from 10.0.1.13: bytes=32 time<1ms TTL=123
Reply from 10.0.1.13: bytes=32 time<1ms TTL=123
Reply from 10.0.1.13: bytes=32 time<1ms TTL=123

Ping statistics for 10.0.1.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.5.0.2

Pinging 10.5.0.2 with 32 bytes of data:

Reply from 10.5.0.2: bytes=32 time=11ms TTL=123
Reply from 10.5.0.2: bytes=32 time<1ms TTL=123
Reply from 10.5.0.2: bytes=32 time=1ms TTL=123
Reply from 10.5.0.2: bytes=32 time=1ms TTL=123

Ping statistics for 10.5.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>

```

Figure 3.14: Successful Inter-VLAN Communication Test

3.2.3 DHCP Verification

DHCP functionality was tested by configuring client devices to obtain their network information automatically. The successful assignment of IP address, subnet mask, default gateway, and DNS information confirms that the DHCP pools are operating correctly.

```

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:BAFF:FESA:4476
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 10.3.0.67
    Subnet Mask . . . . .: 255.255.255.224
    Default Gateway . . . . .: ::
                                10.3.0.65

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

```

Figure 3.15: Successful DHCP IP Assignment

3.2.4 EIGRP Routing Verification

The EIGRP configuration was verified to ensure that routing information was exchanged dynamically between the hospital sites. The routing table and EIGRP neighbor information confirm that the routers are learning and maintaining the required network routes.

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 26 subnets, 6 masks
D    10.0.0.0/24 [90/28416] via 10.255.0.2, 00:40:33, GigabitEthernet0/1
D    10.0.1.0/26 [90/28416] via 10.255.0.2, 00:40:33, GigabitEthernet0/1
D    10.0.1.64/26 [90/28416] via 10.255.0.2, 00:40:33, GigabitEthernet0/1
D    10.0.1.160/28 [90/28416] via 10.255.0.2, 00:40:33, GigabitEthernet0/1
D    10.1.0.0/26 [90/33536] via 200.1.1.2, 00:40:01, GigabitEthernet0/0
D    10.1.0.64/27 [90/33536] via 200.1.1.2, 00:40:01, GigabitEthernet0/0
D    10.1.0.96/28 [90/33536] via 200.1.1.2, 00:40:01, GigabitEthernet0/0
D    10.2.0.0/26 [90/33536] via 200.1.1.2, 00:40:01, GigabitEthernet0/0
D    10.2.0.64/27 [90/33536] via 200.1.1.2, 00:40:01, GigabitEthernet0/0
D    10.2.0.96/28 [90/33536] via 200.1.1.2, 00:40:01, GigabitEthernet0/0
D    10.3.0.0/26 [90/33536] via 200.1.1.2, 00:40:01, GigabitEthernet0/0

Router>
Router>
Router>enable
Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H  Address          Interface    Hold Uptime   SRTT  RTO  Q  Seq
                               (sec)          (ms)  Cnt  Num
0  10.255.0.2        Gig0/1       12   00:40:11   40    1000  0   55
1  200.1.1.2         Gig0/0       11   00:39:39   40    1000  0   55

```

Figure 3.16: EIGRP Routing and Neighbor Verification

3.2.5 DNS Service Test

The DNS service was tested from a client device to verify the resolution of a configured domain name into its corresponding IP address. The client successfully queried the configured DNS server and resolved the domain name, confirming that the DNS service is correctly configured and accessible through the hospital network.

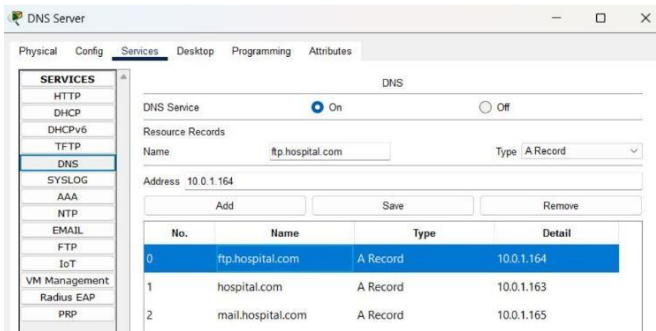


Figure 3.17: Successful DNS Name Resolution Test

3.2.6 HTTP Service Test

The HTTP service was tested from a client device using the Web Browser in Cisco Packet Tracer. Successful access to the configured web server confirms that the HTTP service is active and accessible through the hospital network.

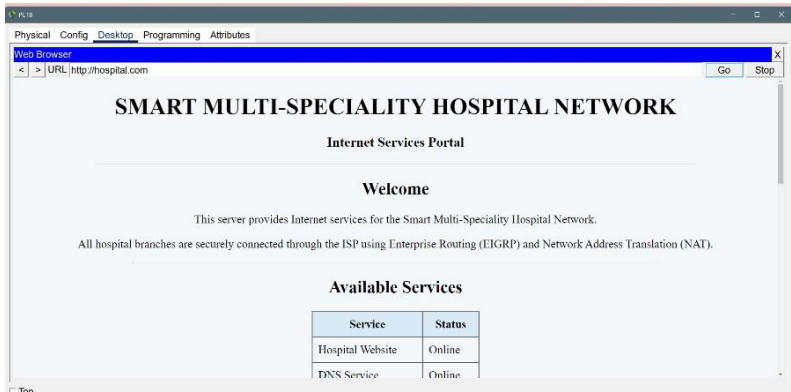


Figure 3.18: Successful HTTP Service Access

3.2.7 FTP Service Test

The FTP service was tested from a client device to verify communication with the configured FTP server. Successful FTP access and file transfer demonstrate that the FTP service is correctly configured and reachable from the client network.

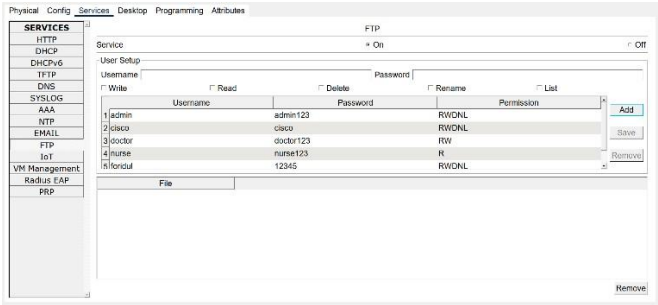
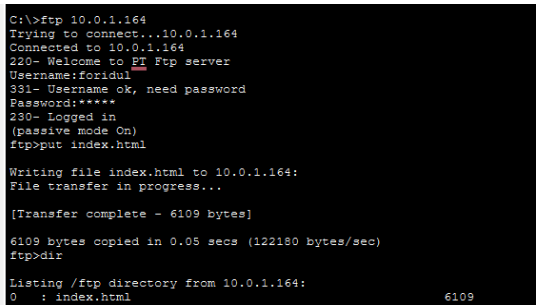


Figure 3.19: Successful FTP Service Access

3.2.8 NAT/PAT Verification

NAT/PAT functionality was tested on the Edge Routers to verify Internet access for internal users. The NAT translation results confirm that internal addresses are translated through the configured overload mechanism while internal branch traffic remains within the private network.

```
Router>
Router>
Router>enable
Router#show running-config | include ip nat
ip nat outside
ip nat inside
ip nat inside source list NAT_ACL interface GigabitEthernet0/0 overload
Router#
Router#
Router#show access-lists NAT_ACL
Extended IP access list NAT_ACL
deny ip 10.2.0.0 0.0.0.255 10.0.0.0 0.255.255.255
permit ip 10.2.0.0 0.0.0.255 any
```

Figure 3.20: NAT/PAT Translation and Internet Connectivity Test

3.2.9 Email Service Test

The email service was tested by configuring user accounts on the hospital mail server and exchanging test messages between client devices. Successful email transmission and reception confirmed that the configured SMTP and POP3 services were active, reachable, and functioning correctly across the hospital network.

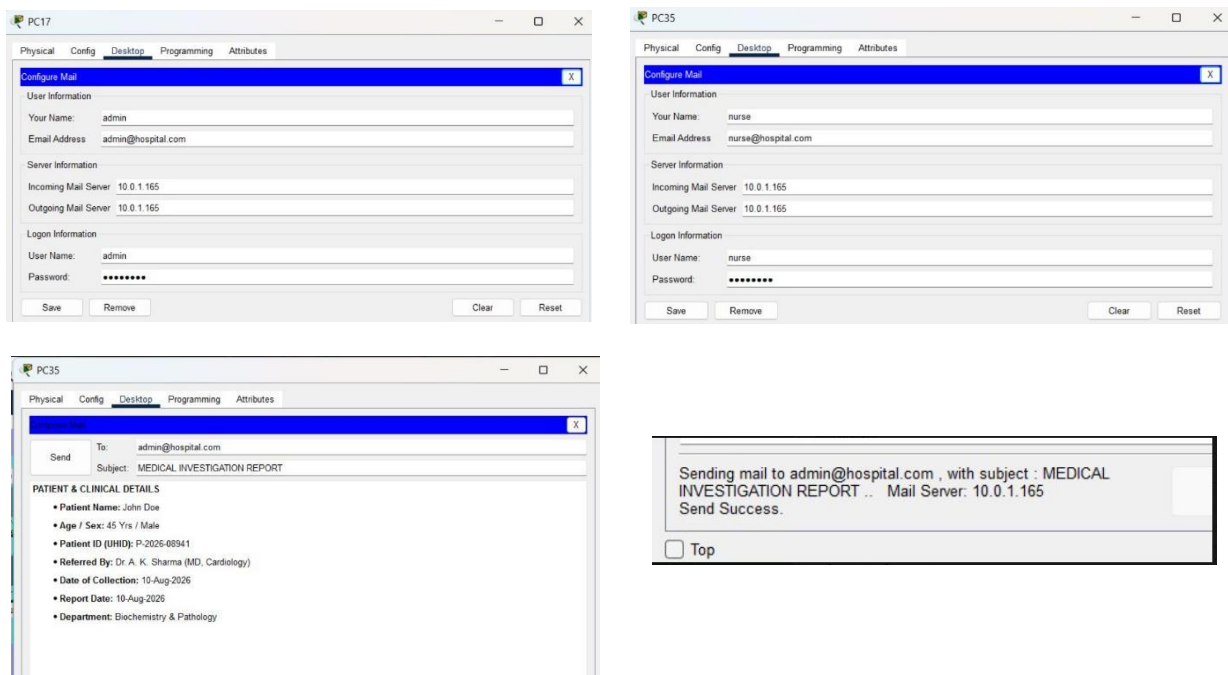


Figure 3.21: Successful Email Service Test

3.2.10 Network Test Cases and Validation Results

This table presents the test cases performed to verify the connectivity, routing, network services, NAT/PAT, and security configurations of the proposed Smart Multi-Speciality Hospital Network. It summarizes the test inputs, references, expected outputs, actual results, and pass/fail status.

Table 3.1: Network Test Cases and Validation Results

Test	Input Description	Ref	Expected Result	Actual Result	Pass/Fail
DHCP Test	Client requests IP automatically	[T1]	Valid IP assigned	IP successfully assigned	Pass
Ping Test	Ping between network devices	[T2]	Successful reply	Reply received	Pass
Inter-VLAN Test	Communication between VLANs	[T3]	Communication allowed	Successful	Pass
EIGRP Test	Verify learned routes	[T4]	Routes available	Routes learned	Pass
Inter-Branch Test	Ping between hospital branches	[T5]	End-to-end reply	Successful	Pass
DNS Test	Resolve configured domain	[T6]	Name resolved	Successfully resolved	Pass
HTTP Test	Access web server	[T7]	Web page loads	Successfully loaded	Pass
FTP Test	Connect to FTP server	[T8]	Server accessible	Successfully connected	Pass
Email Test	Send and receive email	[T9]	Message delivered	Successfully delivered	Pass
NAT/PAT Test	Internal client accesses external network	[T10]	External connectivity	Successfully connected	Pass
End-to-End Test	Test complete network	[T13]	Full connectivity	Successfully verified	Pass

3.3 Discussion

The proposed Smart Multi-Speciality Hospital Network successfully integrates multiple networking technologies to provide reliable communication between the Main Hospital and five branch hospitals. The combination of VLSM, VLAN, Router-on-a-Stick, DHCP, EIGRP, NAT/PAT, ACL, and network services provides a structured and manageable network infrastructure.

VLSM improves IP address utilization, while VLANs provide departmental segmentation and Router-on-a-Stick enables communication between different VLANs. EIGRP provides dynamic route exchange between hospital sites, while DHCP simplifies IP address management. NAT/PAT enables Internet connectivity for internal users, and ACLs provide controlled access to network resources. The implemented network services further support the communication and operational requirements of the hospital environment.

The network was validated using Ping, Traceroute, routing-table verification, NAT/ACL checks, and Packet Tracer Simulation Mode. The successful test results confirm that the proposed design provides **reliable connectivity, efficient routing, controlled access, and scalability** across the hospital network.

Chapter 4

Engineering Standards and Mapping

This chapter discusses the engineering considerations followed throughout the development of the proposed Smart Multi-Speciality Hospital Network. It presents the project's impact on life, society, environment, ethics, and sustainability, followed by the mapping of the project with the required Program Outcomes and complex engineering problem attributes.

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

The proposed network provides reliable communication between the Main Hospital and five specialized hospital branches. It allows different departments and healthcare facilities to communicate and access required network resources through a structured network infrastructure. The use of DHCP, dynamic routing, VLANs, and network services reduces manual network management and supports efficient communication. Reliable connectivity between hospital sites can help staff access required digital services and information more efficiently, contributing to smoother hospital operations.

The network architecture also separates departmental traffic through VLANs and applies ACL-based access control. This provides controlled access to network resources and helps reduce unnecessary or unauthorized communication within the hospital infrastructure.

4.1.2 Impact on Society & Environment

A connected multi-site hospital network can support better communication and resource sharing between healthcare facilities. The proposed architecture allows different hospital branches and departments to communicate through a common network infrastructure, which can improve coordination and accessibility of digital services. From an environmental perspective, the project is initially developed and tested using Cisco Packet Tracer, reducing the need for physical networking equipment during the design and testing stage. This minimizes the consumption of hardware resources during development and allows configuration problems to be identified before potential physical deployment.

The structured network design can also reduce unnecessary duplication of network resources by allowing shared services and centralized management where appropriate.

4.1.3 Ethical Aspects

Ethical considerations are important because a hospital network may carry sensitive organizational and healthcare-related information. The proposed design therefore incorporates security mechanisms such as **VLAN segmentation and Access Control Lists (ACLs)** to control communication and restrict unauthorized access. Network administrators should follow appropriate access-control practices and provide users only with the network access required for their responsibilities. Configuration, testing, and troubleshooting should also be performed responsibly without intentionally disrupting network services or exposing confidential information.

The project emphasizes responsible network design by considering **security, controlled accessibility, and proper resource management** alongside connectivity requirements.

4.1.4 Sustainability Plan

The proposed network is designed to support future expansion without requiring a complete redesign of the existing infrastructure. The use of VLSM allows IP addresses to be allocated according to actual host requirements, while reserved address space can support future users and devices. VLAN segmentation allows additional departments to be incorporated into the network in a structured manner. Similarly, EIGRP dynamic routing allows routing information to be updated automatically as the network grows.

The architecture can therefore be extended by adding new departments, devices, services, or hospital branches while maintaining the existing network structure. This approach supports the long-term scalability, maintainability, efficient resource utilization, and sustainability of the proposed network.

4.2 Project Management and Teamwork

The project was completed through collaborative network design, implementation, configuration, testing, troubleshooting, and documentation. Each team member contributed according to their assigned responsibilities.

Student Name	Role	Work Details
Md. Sultan Mahmud Rasel (241-15-521)	Project Lead & Network Implementation Lead	1. Lead the overall project planning and implementation.
		2. Designed the network topology and architecture.
		3. Configured routers, switches, VLANs, routing, and network services.
		4. Assisted in project report and documentation.
		5. Supervised the overall network integration.
Md. Foridul Islam (241-15-429)	Network Design, IP Configuration & Documentation	1. Contributed to the network design and architecture.
		2. Configured IP addressing and VLSM.
		3. Participated in router and switch implementation.
		4. Performed connectivity testing and troubleshooting.
		5. Prepared and organized the project report and documentation.
Arpita Kundu (241-15-143)	Network Implementation, Testing and Troubleshooting.	1. Network design and implementation.
		2. Assisted with router and switch configuration.
		3. Assisted with VLAN, routing, and network service configuration.
		4. Participated in connectivity testing and troubleshooting.
		5. Assisted in project report and documentation.

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

In this section, provide a mapping of the problem and provided solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

POs	Justification of Mapping (Project Perspective)
PO5	The project incorporates modern network modeling and simulation tools, specifically Cisco Packet Tracer, to design, configure, test, and analyze contemporary network topologies and protocols under realistic conditions before physical deployment.
PO2	The project requires students to identify and analyze complex networking problems by systematically troubleshooting routing protocols, IP addressing, and security configurations within Cisco Packet Tracer. By examining real-time simulation traffic and routing tables, they apply engineering principles to reach substantiated conclusions regarding network convergence and performance.
PO3	This project involves designing and implementing a scalable network infrastructure using efficient routing protocols (such as OSPF, EIGRP, or RIP), variable-length subnet masking (VLSM), and network security configurations (ACLs and NAT). This directly addresses complex engineering problems and network system design challenges by ensuring reliable, safe, and secure data flow

4.3.2 Complex Problem Solving

Knowledge profile and rationale thereof.

Table 4.3: Mapping with complex problem solving.

	EP1	EP2	EP3	EP4	EP5	EP6	EP7
Justification	Dept of Knowledge	Range of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent Extent Stakeholder Involvement	Inter-dependence
	✓	✓	✓				

EP1: This project required the design and implementation of a fully functional network topology from scratch. This involved tackling real-world engineering challenges, such as efficient IP addressing using VLSM, configuring routing protocols for inter-network communication, and securing the system with NAT and ACLs. Working through these roadblocks provided a deeper understanding of how to solve complex network architecture problems.

EP2: Configuring and analyzing network services across varied topologies requires students to balance a range of conflicting requirements, such as performance versus resource constraints. Using Cisco Packet Tracer allows them to systematically test and optimize device configurations to resolve these competing operational demands.

EP3: This project maps to EP3 because troubleshooting complex routing protocols, NAT, and ACLs requires deep analytical investigation of packet flows, routing tables, and security policies to identify and resolve faults. This intensive diagnostic process directly fulfills the "depth of analysis required" to ensure network efficiency and security.

4.3.3 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping, add subsections to put rationale (Use Table 4.3).

Table 4.4: Mapping with complex engineering activities.

Justification	EA1	EA2	EA3	EA4	EA5
	Range of Resources	Level of Interaction	Innovation	Consequences for Society and Environment	Familiarity
	✓	✓	✓		

Chapter 5

Conclusion

5.1 Summary

This project presented the design and implementation of a Smart Multi-Speciality Hospital Network using Cisco Packet Tracer. The proposed network connects the Main Hospital and five specialized branch hospitals through a structured and scalable network architecture. The design was developed to support reliable communication between different departments and hospital sites while maintaining appropriate security, accessibility, and efficient resource utilization. The project applied several core networking technologies, including VLSM-based IP addressing, VLAN segmentation, Router-on-a-Stick inter-VLAN routing, DHCP, EIGRP dynamic routing, NAT/PAT. These technologies were integrated to create a network capable of supporting departmental communication, inter-branch connectivity, Internet access, automatic IP assignment, and controlled network access. The network was implemented and configured in Cisco Packet Tracer, where routers, switches, servers, access points, and end devices were configured according to the proposed architecture. The network was then tested using Ping, Traceroute, routing-table verification, NAT/PAT verification, and Packet Tracer Simulation Mode. The testing results demonstrated successful DHCP assignment, inter-VLAN communication, EIGRP route exchange, inter-branch connectivity, HTTP, FTP, DNS, NAT/PAT, and end-to-end communication. These results indicate that the major components of the proposed network were correctly configured and successfully integrated.

Overall, the project demonstrates how fundamental networking concepts can be combined to develop a reliable, secure, scalable, and manageable multi-site hospital network infrastructure. The completed simulation provides a practical representation of how such a network can be designed, implemented, tested, and analyzed before deployment in a real-world environment.

5.2 Limitation

- The network was developed and tested only in **Cisco Packet Tracer**.
- Real-world hardware performance and physical connectivity were not evaluated.
- Large-scale traffic and network failure scenarios were not extensively tested.
- Advanced security systems such as firewalls and IDS were not implemented.

5.1 Future Work

- Deploy the proposed network in a **real hospital environment**.
- Implement advanced security mechanisms such as **firewalls, VPN, and IDS/IPS**.
- Add network monitoring and centralized management.
- Introduce redundant links and backup devices for higher reliability.
- Integrate secure healthcare applications and services.
- Expand the network to support additional departments and hospital branches.
- Evaluate performance under real-world traffic and failure conditions.

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